

Hamza AFIF

+1(263)384-2216 | ha.afif25@gmail.com | linkedin.com/in/afif-hamza/ | github.com/HamzaAfif | hamzaafif.me



EDUCATION

Université du Québec à Montréal (UQAM)

Master's in Software Engineering

Montreal, Canada

2025 – Present

HEM Engineering School

Bachelor's in Computer Engineering

Casablanca, Morocco

2022 – 2025

PROJECTS

RosaBot | *FastAPI, LangChain, React, Docker, GitHub Actions*

rosa-bot.vercel.app

- Bilingual (FR/EN) AI assistant answering menu, allergen and cocktail-spec questions for staff at a Montreal restaurant. Chose LLM function-calling over structured menu data instead of vector RAG, since allergen and recipe facts are exact and cannot tolerate similarity-based retrieval.
- Built an automated evaluation suite over real staff questions that raised answer accuracy from **44% to 81%**, surfacing a missing menu section and a dairy dish misclassified as vegan. Deployed on Railway/Vercel with GitHub Actions CI.

Clone Consensus | *FastAPI, Server-Sent Events, OpenRouter, Docker*

clone-consensus.up.railway.app

- Treats code clone detection as a consensus problem: a self-implemented deterministic token-based detector runs alongside four LLMs on the same Java source, with a judge model reconciling their verdicts and results streaming live to the browser via SSE.
- Used the classical detector's known Type-1/Type-2-only coverage as a baseline to separate genuine Type-3 clones from model hallucinations.

Student Major Recommender | *scikit-learn, Groq/LLaMA, Node.js*

github.com/HamzaAfif/studentMajorPrediction

- Predicts a student's best-fit academic major from marks and preferences using KNN classification and K-Means clustering (PCA, cross-validation, confusion matrices), with a Groq/LLaMA layer turning model output into readable justifications over a Node.js/Express and Socket.IO interface.

EasyFind | *Python, GPT-4o-mini, LangChain, Telegram*

Prix Coup de Cœur, GSEC 2024, Collège LaSalle

- AI commerce assistant letting non-technical sellers list and manage products entirely through natural-language Telegram conversation (LangChain NL-to-SQL over SQLite), with location-aware discovery for buyers. Won the Moroccan national final, then the Prix Coup de Cœur at GSEC 2024 in Montreal.

AMINA Refactored | *Python, design patterns, pytest*

github.com/HamzaAfif/amina-refactored

- Restructured a handwriting-input math tutor into a layered pipeline architecture using five design patterns (Chain of Responsibility, Strategy, Factory Method, Decorator, Adapter), with 52 regression tests proving the refactoring preserved behaviour end to end — live: amina-refactored.up.railway.app.

EXPERIENCE

Cybersecurity Team Lead (Internship)

Jul 2024 – Oct 2024

Microtech Leaders (MSSP)

Casablanca, Morocco

- Led a team of interns through client security operations; deployed ConnectWise MDR and SentinelOne agents across client endpoints.
- Built a log-analysis tool parsing raw security events into structured incident reports, replacing a manual triage step, and a Django platform delivering personalized training modules.

IT and Automation Intern

Jan 2025 – Feb 2025

Centrelec

Casablanca, Morocco

- Built internal automation on a self-hosted LLaMA model to process routine documentation tasks; maintained and troubleshot network infrastructure across the site.

Software Development Intern

Jun 2024 – Jul 2024

Moroccan Jewish Museum

Casablanca, Morocco

- Built Python/Flask tools for a museum running entirely on paper records: a Telegram bot delivering QR-triggered audio storytelling along the exhibition route, plus a QR-based inventory system for the collection.

TECHNICAL SKILLS & CERTIFICATIONS

Languages: Python, JavaScript, Java, C/C++, PHP, SQL, Bash

Backend & Web: FastAPI, Flask, Django, Spring Boot, Laravel, Node.js/Express, React, Angular, PostgreSQL, MongoDB

AI & ML: LangChain, OpenAI/Groq APIs, LLM function-calling, NL-to-SQL, LangSmith, scikit-learn (KNN, K-Means, PCA)

Infrastructure: Docker, GitHub Actions, AWS EC2, NGINX, Railway, Vercel, Cloudflare Pages, Linux

Testing & Quality: pytest, Gatling, JMeter, SonarQube/SonarCloud, CodeScene, PMD/CPD

Certifications: AWS Developer Associate (DVA-C02, in progress), Cisco CCNA (ITN), Cisco IT Essentials

Languages: French — DALF C1 | English — Linguaskill Business C1 | Arabic — native